

TEZPUR UNIVERSITY
Type A Test III (Major I), Autumn 2013
CO 504 / CS 537 NATURAL LANGUAGE
PROCESSING

Marks : 25

Time : 1 hour

The figures in the right-hand margin indicate marks for the individual question.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

1. Give an example of “free word order” in a natural language. 2
2. (a) Why is morphological analysis required? 3
 (b) Give two examples of regular expressions and draw the state transition diagram for the corresponding finite automata. 3
 (c) Give an example of Finite State Transducer based morphological lexicon entry. 1
3. Give an example to show how the concept of n-gram can be used in natural language processing. 2
4. Perform PoS tagging for the following sentence-
I would have loved to shop in Fancy Bazar.
 Indicate which words are ambiguous in PoS and suggest a rule that be used in rule based PoS tagging of the above sentence. 2 + 2 + 3

5. (a) Write a context free grammar for simple English sentences with some left recursive rules. 3

(b) How does left recursion create problem in top down parsing? How can left recursion in your CFG above be eliminated? 2 + 2